

# World Cup 2026 Prediction App Summary

*Repo-based one-page brief*

## What It Is

Notebook-driven machine learning project for predicting football match outcomes (home win, draw, away win) and using those predictions to simulate the 2026 FIFA World Cup. The README describes it as an early-stage baseline predictive framework, not a final production system.

## Who It's For

Not found in repo.

## What It Does

- ? Cleans and standardizes multiple tournament datasets into a shared match schema.
- ? Combines match history, Elo-style ratings, xG, yellow cards, and qualification status.
- ? Builds engineered team-level features and train/test datasets for modeling.
- ? Compares baseline and ML classifiers including Logistic Regression, Random Forest, XGBoost, Gradient Boosting, and SVM.
- ? Evaluates models with accuracy, classification reports, and confusion matrices.
- ? Saves the selected model, scaler, and feature list for reuse in prediction.
- ? Runs Monte Carlo World Cup simulations and exports tournament probability results.

## How It Works

- ? **Inputs:** [data/] CSV/XLSX files for 2022 World Cup matches, 2024 regional tournaments, 2025 Concacaf data, Elo ratings, qualifiers, and team stats.
- ? **Prep:** [notebooks/Data\_Prep.ipynb] reads raw files, standardizes columns, and writes [data/final\_dataset.csv].
- ? **Features:** [notebooks/Feature\_engineering.ipynb] derives team stats and writes [data/data\_features.csv], train/test splits, and [data/team\_stats.csv].
- ? **Modeling:** [notebooks/model\_dev.ipynb] scales features, compares classifiers, and saves [model/best\_model.pkl], [model/scaler.pkl], and [model/feature\_names.pkl].
- ? **Simulation:** [notebooks/Simulation.ipynb] loads saved artifacts, predicts match outcomes, simulates tournament rounds, and writes [results/wc2026\_results.csv].
- ? **Services/UI:** Not found in repo.

## How To Run

- ? Install notebook-used Python packages: pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost, and joblib. Dependency file: Not found in repo.
- ? Open the notebooks and run them in sequence: Data\_Prep, EDA, Feature\_engineering, model\_dev, then Simulation.
- ? Keep the notebook file paths aligned with the repo folders. Notebook references use ../DataSets and ../Models, while this repo stores those assets in [data/] and [model/].
- ? Check exported outputs in [model/] and [results/wc2026\_results.csv]. Entry-point script or packaged app runner: Not found in repo.

## Evidence Base

README, notebook files, saved model artifacts, dataset files, and exported results present in the repo.